The Limits of Verifiability: Credibility and Flexibility in Communication

Alessandro Lizzeri — Princeton
Yichuan Lou — Tokyo
Jacopo Perego — Columbia

August 2026

Motivation

Large variation in how institutions govern communications from experts to decision makers:

- Some orgs require managers to report through standardized performance metrics, others rely more heavily on informal assessments
- Audited vs nonaudited statements in private firms

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Audited vs nonaudited statements in private firms

This variation is partly a matter of institutional design:

- $\sim$ Procedures for collecting evidence and restrict communication to claims supported by that evidence $\rightsquigarrow$ communication is **verifiable**
- $\sim$ Absent these procedures $\rightsquigarrow$ communication is **unverifiable**

Motivation

Verifiable communication is commonly viewed as superior to unverifiable communication, in the sense that it facilitates information transmission

- In many settings, it induces full disclosure, or substantially alleviates info asymmetries (Milgrom 81, Grossman 81, Seidmann and Winter 97...)
- Verifiability, like commitment, helps sender overcome the credibility problems of cheap talk (Sobel 13)
- This perspective is perhaps even more widespread in Accounting (IASB 18)

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This paper qualifies this perspective

This Paper

We compare verifiable and unverifiable communication in sender-receiver settings

Two simple but key features:

- ∼ Sender's evidence is a noisy signal of her private information
- ∼ Sender and receiver have partially aligned interests (à la CS'82)

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- Verifiability is valuable bc restrictions on sender's strategy discipline her claims and enhance her credibility
- Noisy evidence may misrepresent what sender knows. Verifiability restricts sender's flexibility to convey what evidence misses

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2. Verifiable *versus* Unverifiable. When preferences are

- ∼ Sufficiently misaligned ⇝ verifiability facilitates communication
- ∼ Sufficiently aligned ⇝ verifiability hinders communication (design vs technology)

3. Verifiable *plus* unverifiable

- ∼ Study complementarities: unverifiable communication "contextualizes" evidence when the latter is misleading

Related Literature

Conditions for unraveling and its failures Grossman 81, Milgrom 81, Dye 85, Okuno-Fujiwara et al. 90, Seidmann and Winter 97, Mathis 08, Giovannoni and Seidmann 07, Hagenbach et al. 14 [..........]

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Combination of hard and soft: Eso Galambos 13, Bertomeu and Marinovic 16, Dasgupta 23

model

Model

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(S, π) is an **evidence structure** where

- $S = \{s_1, \ldots, s_N\}$ and $s_1 < \ldots < s_N$
- $\pi : \Theta \to \Delta(S)$

(S, π) satisfies "Full support" $\rightsquigarrow \pi(s \mid \theta) > 0$ almost everywhere in $\Theta \times S$

MLRP (for expositional convenience)

We say (S, π) is **informative** if there is (s, s') s.t. $F_s \neq F_{s'}$

Model

Timing:

1. A **sender** privately observes (θ, s), her "type"
2. She sends a message $m \in M$ to **receiver**, where $M \triangleq S \cup \{\circ\}$
3. Receiver observes m, updates her belief about θ, and takes action $a \in \mathbb{R}$

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We study this model under two alternative communication technologies

Verifiable evidence:

- Sender of type (θ, s) can either announce s or be silent, $m \in \{s, \circ\}$
- Sender's strategy is **constrained**

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Verifiable evidence:

- Sender of type (θ, s) can either announce s or be silent, $m \in \{s, \circ\}$
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Unverifiable evidence:

- Sender of type (θ, s) can announce any s' or be silent, $m \in S \cup \{\circ\}$
- Sender's strategy is **unconstrained**

Model

Sender-receiver payoffs as in Crawford and Sobel 82

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There is differentiable $u : A \times \Theta \times \mathbb{R}_+ \to \mathbb{R}$ s.t.

Payoffs: $u^{\mathcal{R}}(a, \theta) \triangleq u(a, \theta, 0), \quad u^{\mathcal{S}}(a, \theta, b) \triangleq u(a, \theta, b), \quad b \geq 0$ sender's **bias**

Preferred Actions: $a^{\mathcal{R}}(\theta), \quad a^{\mathcal{S}}(\theta, b)$

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Assumptions on u:

- $\sim$ $a^{\mathcal{R}}(0) = 0, \quad a^{\mathcal{R}}(1) = 1 \rightsquigarrow$ normalization
- $\sim$ $u_{11} < 0 \rightsquigarrow$ unique preferred actions
- $\sim$ $u_{12} > 0 \rightsquigarrow$ higher θ prefers higher a
- $\sim$ $u_{13} > 0 \rightsquigarrow$ higher b prefers higher $a \rightsquigarrow a^{\mathcal{R}}(\theta) \leq a^{\mathcal{S}}(\theta, b)$
- $\sim$ $\lim_{b \to \infty} u_1(a, \theta, b) > 0 \rightsquigarrow$ for large bias, sender always prefers higher a

Example

To illustrate, I will use this example throughout the talk

F is the uniform CDF on $[0,1]$

Evidence structure (S,π):

- $S = \{s_L, s_H\}$
- $\pi(s_H \mid \theta) = \theta$ and $\pi(s_L \mid \theta) = 1-\theta$

Payoffs

- $u^{\mathcal{R}}(a,\theta) = -(a-\theta)^2$ $\qquad a^{\mathcal{R}}(\theta) = \theta$
- $u^{\mathcal{S}}(a,\theta) = -(a-\theta-b)^2$ $\qquad a^{\mathcal{S}}(\theta) = \theta + b$

discussion

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The *combination* of these two features drives our results

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Many disclosure environments share these two features:

- Manager knows more about performance than formal metrics capture
- Loan officer knows more about borrower quality than financial history reveals
- An expert witness knows more than admissible evidence can convey

In all these cases,

- Sender knows more than what her evidence can prove
- Sender and receiver may be only partially aligned

More Discussion

We see the two games as **polar** communication protocols

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- $\sim$ Certification is unavailable or prohibitively costly

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- In our setting, shutting down cheap talk can be with loss
- Institutional commitment to base decisions only on verifiable information
- Milgrom and Roberts '88: organizations shut down communication channels to limit wasteful influence activities

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Later, we will study a game with both protocols available at the same time

outcomes and welfare

Some Basic Terminology

Let an **outcome** be a joint distribution of action and state, $\lambda \in \Delta(A \times \Theta)$

We say outcome is **uninformative** if action (a) is independent of state (θ).

$\sim$ Otherwise, it is **informative**

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Definition

Let λ and λ' be two outcomes. We say that λ is **more informative** than λ' if $U(\lambda) \geq U(\lambda')$.

We call the most-informative outcome **receiver-optimal**

communicating with verifiable evidence

Communicating with Verifiable Evidence

We begin by analyzing the game with **verifiable evidence**

Unlike in typical disclosure games, the set of equilibria is quite rich

Its characterization reveals the forces shaping information transmission in our setting

Full Evidence Disclosure

Natural point of departure is to ask whether an analogue of the unraveling argument holds in our setting

- I.e., does equilibrium exist where receiver learns the evidence s (i.e., the verifiable content of sender's private info)?

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Let's give such an outcome a name:

Definition

Full-evidence-disclosure (FED) is the outcome induced when sender always discloses her realized signal and receiver best responds

A **FED equilibrium** is any equilibrium that induces this outcome

When Is FED Possible?

Proposition 1

Consider the game with verifiable evidence. Suppose evidence is informative.

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The **positive part** is in line with the disclosure literature: when the sender is very biased, there is an off-path punishment so severe that all types prefer to disclose

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The **negative part** is novel: when sender and receiver are sufficiently aligned, there is no eqm in which receiver learns the signal realization

Intuition

To understand this failure, consider extreme case: $b = 0$ (aligned pref)

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Consider a sender's type $(\theta_\circ, s)$, where $a^{\mathcal{R}}(\theta_\circ) = a_\circ$:

- $\sim$ Stick to eqm and send $m = s$, thus inducing action $a^{\mathcal{R}}(F_s)$ OR
- $\sim$ Deviate to $m = \circ$ and induce $a^{\mathcal{R}}(\theta_\circ) = a_\circ$

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- $\sim$ Deviate to $m = \circ$ and induce $a^{\mathcal{R}}(\theta_\circ) = a_\circ$

Since evidence is informative, $\exists \tilde{s}$ such that $a^{\mathcal{R}}(F_{\tilde{s}}) \neq a_\circ$, $\rightsquigarrow$ strict profit deviation

Interpretation

Iff preferences are sufficiently aligned, full evidence disclosure (FED) is impossible

The proof intuition reflects a more general lesson:

- There is no way to assign an off-path interpretation to silence that is uninformly unattractive to **all** sender's types
- Any off-path action will attract some sender type, providing them with the incentives to **separate** from other types

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This "separating force" suggests sender may be able to transmit **more** information than that encoded in her verifiable evidence

The Value of Silence

Proposition 2

Consider the game with verifiable evidence.

1. If $b \geq \bar{B}$, receiver-optimal equilibria are as informative as the FED outcome

2a. If b is sufficiently small, receiver-optimal equilibria are strictly more informative than FED outcome

2b. For every $b < \bar{B}$, there exists F and (S, π) s.t. receiver-optimal equilibria are strictly more informative than FED

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Thus, when objectives are sufficiently aligned, <u>FED is not only impossible, it is also inefficient</u>

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Policy implication: when sender's bias is small, mandating disclosure can hurt receiver

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Note importance of both key assumptions: (1) partially aligned preferences and (2) sender knowing more than her evidence can prove $\rightsquigarrow (\theta, s)$

Example (continued)

Let's go back to example:

- $\sim$ uniform prior, quadratic payoffs, $b = 0$, binary and "linear" signals

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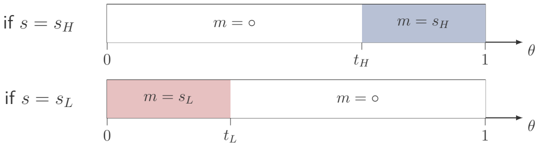

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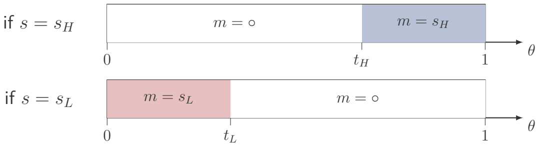

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When b is small, such a strategy is eqm and more informative than FED outcome. It also induces the receiver-optimal outcome (comp stats)

verifiable vs unverifiable communication

Verifiable vs Unverifiable Communication

When evidence is verifiable, silence can facilitate information transmission because it can be used flexibly

What happens when the sender has unrestricted flexibility over the message she sends?

Verifiable vs Unverifiable Communication

When evidence is verifiable, silence can facilitate information transmission because it can be used flexibly

What happens when the sender has unrestricted flexibility over the message she sends?

Next, we compare receiver-optimal equilibria under verifiable and unverifiable communication

recall, same message space

- Which regime induces the most informative equilibria?

  Note: in standard disclosure models, this question is moot: verifiability obviously dominates, since receiver fully learns θ

Verifiable vs Unverifiable Communication

Proposition

There exist thresholds $0 < \beta_0 < \beta_1 < \beta_2 \leq \infty$ such that:

1. For all $b \leq \beta_0$, receiver-optimal equilibria of the game with verifiable evidence are **strictly less informative** than those of the game with unverifiable evidence

2. For all $b \in (\beta_1, \beta_2)$, receiver-optimal equilibria of the game with verifiable evidence are **strictly more informative** than those of the game with unverifiable evidence.

   Moreover, if evidence structure is informative, $\beta_2 = \infty$, so the comparison holds for all $b > \beta_1$

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Focus first on case of informative evidence

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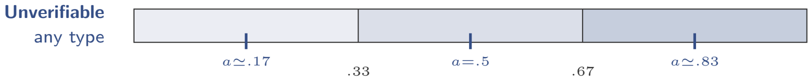

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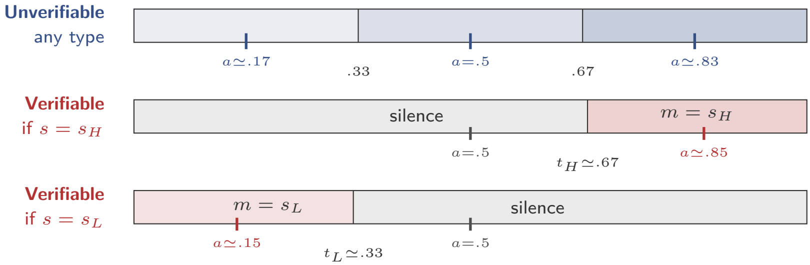

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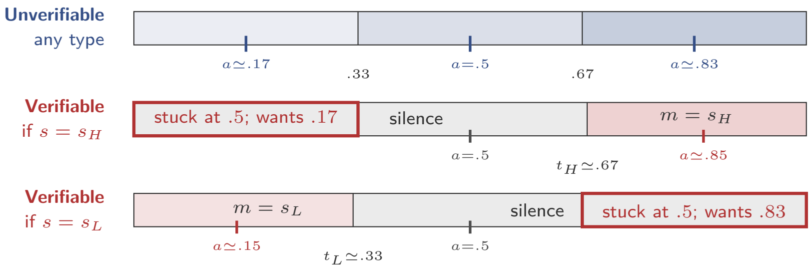

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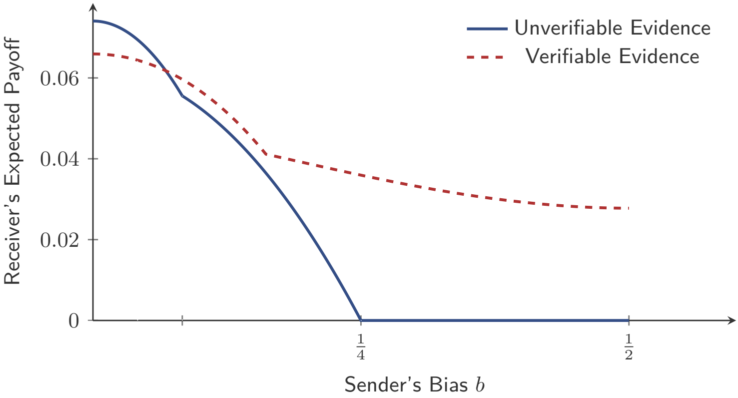

Unverifiable Evidence
Verifiable Evidence
Receiver's Expected Payoff
0.06
0.04
0.02
0
$\frac{1}{4}$
$\frac{1}{2}$
Sender's Bias b

Interpretation: Flexibility vs Credibility

When b is small, **flexibility** is valuable:

- $\sim$ Verifiable communication constrains the sender, $m \in \{\circ, s\}$
- $\sim$ Unverifiable communication lets her flexibly communicate her private info

Interpretation: Flexibility vs Credibility

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When b is large, **credibility** is valuable:

- $\sim$ Unverifiable communication is too flexible $\rightsquigarrow$ imitation
- $\sim$ Verifiable communication disciplines sender and restores credibility

Interpretation

- Verifiability facilitates communication when the bias is large, but hinders it when the bias is small.
- This result qualifies the common view that verifiability necessarily improves information transmission
- Even when certification is feasible, verifiability may not be desirable

  design vs technology
- Same message space: result driven only by the implicit constraints on sender's strategy imposed by verifiability

Empirical Applications

An application of our framework is in **internal organizational reporting**

- ∼ How organizations design their internal reporting systems

Lower-level managers often know more than formal metrics capture about

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- Employee performance, customer relationships, operational problems

This info may be conveyed through reporting systems that are more or less

- Standardized and formal: e.g., KPIs and other hard metrics
- Flexible and informal: qualitative updates and subjective assessments

Empirical Applications

An application of our framework is in **internal organizational reporting**

- $\sim$ How organizations design their internal reporting systems

Lower-level managers often know more than formal metrics capture about

- $\sim$ Employee performance, customer relationships, operational problems

This info may be conveyed through reporting systems that are more or less

- $\sim$ Standardized and formal: e.g., KPIs and other hard metrics
- $\sim$ Flexible and informal: qualitative updates and subjective assessments

Reporting systems therefore face the same tradeoff as in our model $\rightsquigarrow$ prediction

Empirical Applications

Empirical evidence seems consistent with this tradeoff:

- Greater hierarchical distance btw loan officers and their supervisors who approve loans reduces reliance on soft info and increases it on hard info
  Liberti and Mian '09 (*RFS*)
- Geographic proximity facilitates the use of soft information
  Petersen and Rajan '02 (*JF*), Agarwal and Hauswald '10 (*RFS*)
- Longer working relationships made decisions more responsive to soft information
  Qian et al '15 (*JF*)

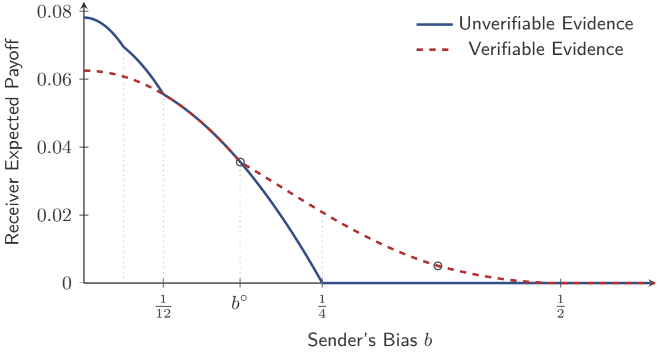
Uninformative Evidence

So far we focused on the case of **informative evidence**

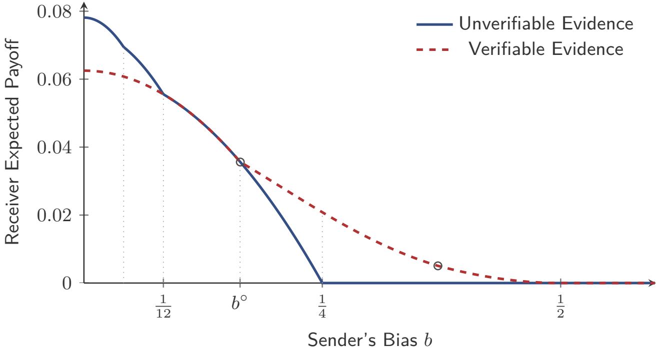
The advantage of verifiability for large b may reflect not just the credibility coming from restricting sender strategy, but the fact that evidence is informative on its own

Yet, our result shows verifiability can improve communication even when evidence is uninformative, purely by restricting which messages each sender type can send

Uninformative Evidence



Uninformative Evidence

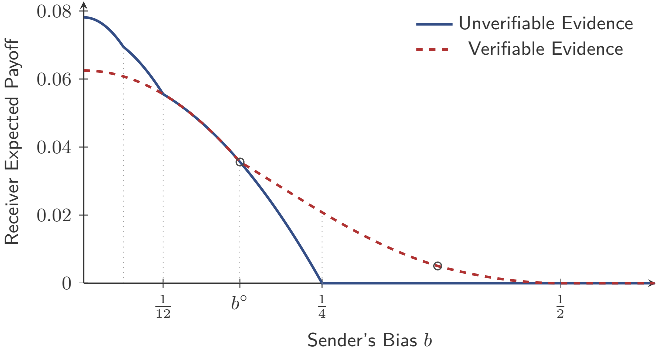


Why? The stochastic availability of uninformative evidence disciplines the sender

This relates to the familiar idea that noise in cheap talk can be beneficial

Myerson, 1991; Krishna and Morgan, 2004; Blume et al., 2007; Goltsman et al., 2009

Uninformative Evidence



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More-Informative Evidence

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More-Informative Evidence

But then, does the informativeness of the evidence play any role at all?

We ask: If the evidence becomes more informative, does this benefit communication? $\rightsquigarrow$ Not necessarily

Remark 1

Suppose (S, π) is more Blackwell-informative than (S', π').

- If $b = 0$ or $b \geq 1$, the efficient equilibrium under (S, π) is more informative than the efficient equilibrium under (S', π')
- For intermediate values of b, the opposite can happen

verifiable *plus* unverifiable communication

Interacting Verifiable and Unverifiable Communication

So far, we have compared the two communication regimes in isolation

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In many applications, communication often combines both

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In many applications, communication often combines both

Standard disclosure models leave no role for interactions with cheap talk: verifiability reveals the sender's private information

Our model provides a natural setting to study their **interactions**:

- $\sim$ Cheap talk can be used to *qualify* misleading evidence

The "Hybrid" Game

Model is as before, except now sender sends <u>two</u> messages, m^V and m^U

- $\sim$ $m^V \in S \cup \{\circ\}$ is **verifiable**: type (θ, s) can choose only $m^V \in \{s, \circ\}$
- $\sim$ $m^U \in M^U$ is **unverifiable**: type (θ, s) can send any message in M^U

To illustrate interactions between verifiable and unverifiable communication, we focus on simple class of equilibria:

"**Reveal-and-Refine**" equilibria:

Sender always discloses her evidence, i.e., $m^V = s$, and then uses m^U to refine the receiver interim posterior

Reveal and Refine

To illustrate interactions between verifiable and unverifiable communication, we focus on simple class of equilibria:

> "**Reveal-and-Refine**" equilibria:
>
> Sender always discloses her evidence, i.e., $m^V = s$, and then uses m^U to refine the receiver interim posterior

This class is useful because

- ∼ Natural analogue of FED in the hybrid game
- ∼ Captures what would happen under **mandatory disclosure** if cheap talk remains possible

Example

Back to example:

- Uniform prior, quadratic payoffs, $b = \frac{1}{5}$, binary and “linear” signals

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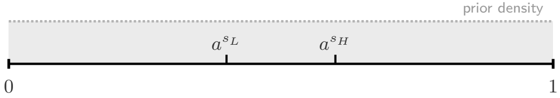
- Uniform prior, quadratic payoffs, $b = \frac{1}{5}$, binary and "linear" signals

Two preliminary observations:

- With verifiable evidence only, FED is not an eqm since $b = \frac{1}{5} < \frac{3}{10} = \bar{b}$
- With unverifiable evidence only, receiver-optimal eqm is a two-cell partition

Example

FED outcome



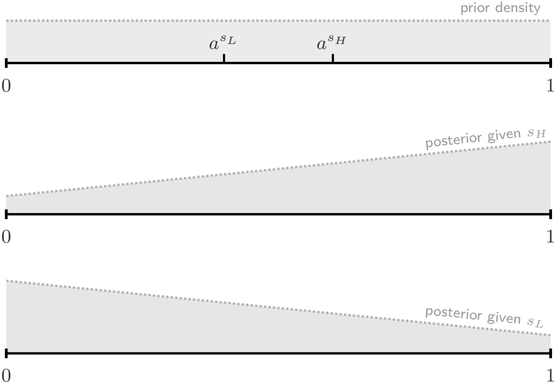
Example

FED outcome

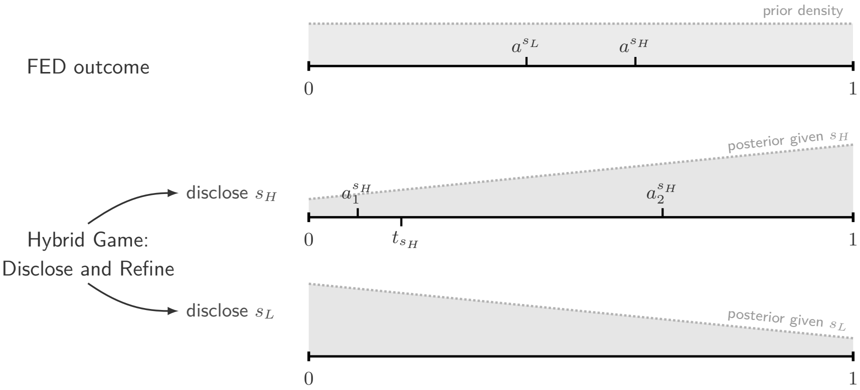
Hybrid Game:
Disclose and Refine

disclose s_H

disclose s_L

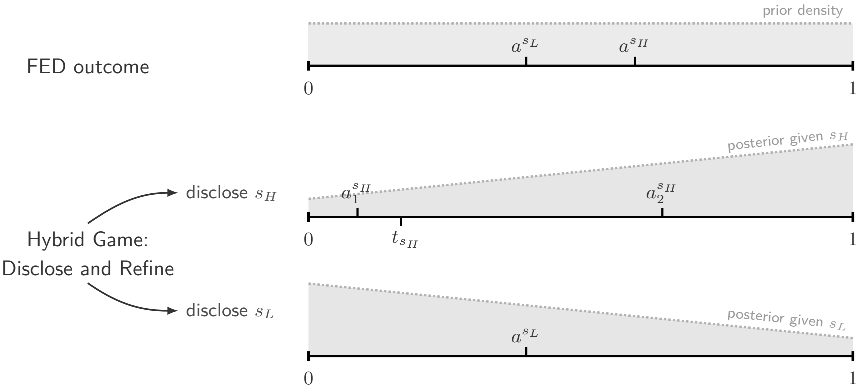


Example



FED outcome
prior density
a^{s_L}
a^{s_H}
0
1
Hybrid Game:
Disclose and Refine
disclose s_H
disclose s_L
posterior given s_H
$a_1^{s_H}$
$a_2^{s_H}$
t_{s_H}
0
1
posterior given s_L
0
1

Example


prior density
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t_{s_H}
1
disclose s_L
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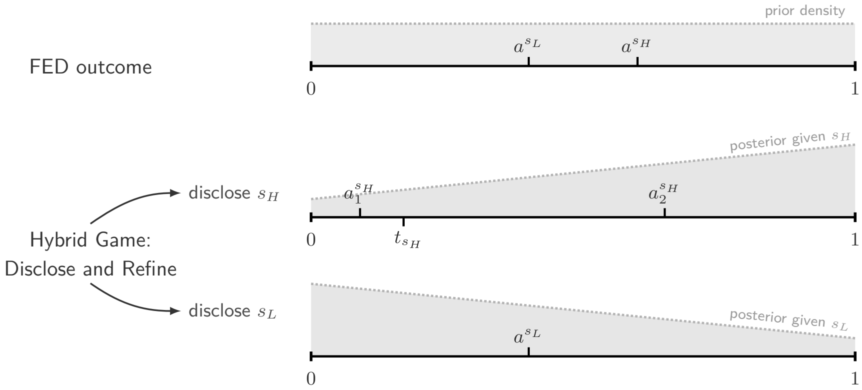
0
1

Example



A disclose-and-refine equilibrium exists

Interacting Verifiable and Unverifiable Communication

Sender discloses her evidence and then uses cheap talk to qualify its interpretation

Interactions btw verifiable and unverifiable change way in which both are used

- Without verifiable, sender would be able to always induce two actions. Yet, with verifiable, she can't
- Without unverifiable, sender would not be able to induce FED. Yet, with unverifiable communication, she can

Interacting Verifiable and Unverifiable Communication

Sender discloses her evidence and then uses cheap talk to qualify its interpretation

Interactions btw verifiable and unverifiable change way in which both are used

- ∼ Without verifiable, sender would be able to always induce two actions. Yet, with verifiable, she can't
- ∼ Without unverifiable, sender would not be able to induce FED. Yet, with unverifiable communication, she can

This is not a special feature of the example. It holds generally:

Proposition

Disclose-and-refine equilibria exist <u>for all</u> $b \geq 0$.

Contrast with Proposition 1

conclusion

Conclusion

We study a standard communication model whose innovation consists of combining two simple features:

- The sender's evidence is a noisy signal of her private information
- The sender and receiver have partially aligned objectives

We use this model to qualify a central lesson of the disclosure literature: verifiability is not always a positive force in communication

We identify a **credibility-flexibility tradeoff** and study its implications

Verifiability facilitates communication when objectives are sufficiently misaligned, but hinders communication when they are sufficiently aligned

Showcase complementarities between verifiable and unverifiable communication

The Limits of Verifiability: Credibility and Flexibility in Communication

Alessandro Lizzeri — Princeton
Yichuan Lou — Tokyo
Jacopo Perego — Columbia

Thank You!